

Smooth source, unstructured head

A flow-matching stress test for time series forecasting

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Does a smooth starting curve need a temporally structured network?

Flow matching turns a random starting curve into a forecast through learned updates guided by past observations.⁴
Hypothesis: with smooth starting curves and targets, an MLP may learn the overall trend but produce unwanted local fluctuations.

01 Evidence and open question

	TimeFlow (Sundial) ¹	TSFlow ²
Source	Independent noise	Correlated GP
Head	AdaLN MLP	DiffWave + S4 ³
History	Transformer state	Past + mask
Forecast	Multi-patch	Fixed windows

What the papers evaluate

Sundial. Zero-shot GIFT-Eval: 23 datasets, 97 settings, 100 sample paths. MASE and CRPS assess point and probabilistic forecasts. Test datasets are excluded from pretraining.¹

TSFlow. Eight univariate datasets. Forecast CRPS uses nine quantiles and 100 samples, over five seeds. Prior comparisons keep the structured head fixed.²

What remains unresolved

Published evidence supports probabilistic forecasting with each chosen source–head pairing.^{1,2} It does not show that a GP prior helps an MLP. Our shared-data 2×2 test separates source choice from architecture.

Planned outputs

Four-cell table and whitening control: PC1 velocity MSE; PCs 3– F velocity MSE; sample-path roughness. Generated paths plotted against targets.

02 The decisive experiment

Same smooth targets and tiny lookback encoder

Source	AdaLN MLP Fully connected network	S4 head Temporal structure, 3 S4 blocks
White Gaussian 	White noise + MLP TimeFlow-style baseline	White noise + S4 Structured network baseline
Matched GP 	GP + MLP Main stress test	GP + S4 TSFlow-style baseline

Experimental setup

Data. Joint smooth GP windows, $F = 16$; matched source kernel and Sundial instance normalization.
Training. Independent source–target pairs in observation space; small models trained from scratch with comparable sizes and budgets.
Evaluation. Independent train/validation/test draws; train-only PCA; a fixed number of Euler steps.

Validate the stress-test data

Fit PCA on training draws; check on validation draws whether the first two PCs capture most velocity variation.

$$E_2 = \frac{\text{variance in the first two PCs}}{\text{total velocity variance}} > 0.90$$

Require this threshold for matched-GP velocities. Report target and both velocity spectra. If it fails, adjust the GP lengthscale; reject constant data.

03 Evaluation and conclusions

Accuracy and smoothness

Compare mean squared error (MSE) of velocity in the dominant pattern (PC1) and weaker patterns (PCs 3, . . . , F). Report PC2 separately.

$$R(x) = \frac{1}{F-1} \sum_{i=1}^{F-1} |x_{i+1} - x_i|$$

$R(x)$ measures adjacent jumps. Generated curves should match target roughness; excessive smoothness also fails.
Proposed success rule: residual error within 20% of PC1 error; roughness within 20% of target. Fix MSE normalization before training.

Does the coordinate scaling matter?

Add a **PCA-whitened MLP** with the same GP source and encoder. Rescale components to equal variance using a transform fitted on training velocities. Evaluate all three metrics back in observation space.

Which head should we keep?

If **whitening closes the gap**, use better-scaled coordinates. If **S4 still wins**, investigate temporal structure. If **MLP matches S4 on all three metrics with a GP source**, the predicted gap is absent. This guides the next structured-prior model, within this synthetic setting.

WEEK 1 Generate synthetic GP data and check PCA spectra

WEEK 2 Implement MLP and S4; run the 2×2 experiments

WEEK 3 Evaluate errors and roughness; test the whitened MLP

WEEK 4 Analyse results, plot paths and write recommendations

[1] Liu et al. *Sundial*, ICML 2025. [arXiv:2502.00816](#).
[2] Kolloviah et al. *Flow Matching with GP Priors*. ICLR 2025. [arXiv:2410.03024](#).

[3] Gu, Goel & Ré. *Structured State Spaces (S4)*. ICLR 2022. [arXiv:2111.00396](#).
[4] Lipman et al. *Flow Matching for Generative Modeling*. ICLR 2023. [arXiv:2210.02747](#).